

Qn	Solution
1 [4]	Since θ is sufficiently small, $\sin \theta \approx \theta$. <u>Method 1: Binomial Series (MF27)</u> $f(\theta) \approx \frac{1}{3-\theta}$ $= (3-\theta)^{-1}$ $= \left[3 \left(1 - \frac{1}{3}\theta \right) \right]^{-1}$ $= 3^{-1} \left[1 + \left(-\frac{1}{3}\theta \right) \right]^{-1}$ $= \frac{1}{3} \left[1 + \left(-\frac{1}{3}\theta \right) \right]^{-1}$ $= \frac{1}{3} \left[1 + (-1) \left(-\frac{1}{3}\theta \right) + \frac{(-1)(-2)}{2!} \left(-\frac{1}{3}\theta \right)^2 + \dots \right]$ $= \frac{1}{3} \left(1 + \frac{1}{3}\theta + \frac{1}{9}\theta^2 + \dots \right)$ $\approx \frac{1}{3} + \frac{1}{9}\theta + \frac{1}{27}\theta^2 \text{ (shown)}$ $\therefore p = \frac{1}{3}, q = \frac{1}{9} \text{ and } r = \frac{1}{27}.$ <u>Method 2: Repeated Differentiation</u> $f(\theta) \approx (3-\theta)^{-1}$ $f'(\theta) \approx -(3-\theta)^{-2}(-1) = (3-\theta)^{-2}$ $f''(\theta) \approx -2(3-\theta)^{-3}(-1) = 2(3-\theta)^{-3}$ Sub $\theta = 0$: $f(0) = 3^{-1} = \frac{1}{3}$ $f'(0) = (3-0)^{-2} = \frac{1}{9}$ $f''(0) = 2(3-0)^{-3} = \frac{2}{27}$ Using MF27, $f(\theta) \approx \frac{1}{3} + \frac{1}{9}\theta + \frac{\frac{2}{27}}{2!}\theta^2 + \dots$ $f(\theta) \approx \frac{1}{3} + \frac{1}{9}\theta + \frac{1}{27}\theta^2 + \dots \text{ (shown)}$ $\therefore p = \frac{1}{3}, q = \frac{1}{9} \text{ and } r = \frac{1}{27}.$

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Qn	Solution
2(i) [3]	From the question, the equation of the curve is $y = \frac{x^2 + px + q}{x - 2}$. Since the vertical asymptote is $x = 2 \Rightarrow r = 2$. C passes through the point $(0, -1.5)$. $y = \frac{x^2 + px + q}{x - 2} \Rightarrow -1.5 = \frac{0 + 0 + q}{0 - 2} \Rightarrow q = 3$ Hence, the equation of C is $y = \frac{x^2 + px + 3}{x - 2}$ ----- (1) C has asymptotes with equations $y = x - 1$ and $x = 2$. $\Rightarrow y = x - 1 + \frac{a}{x - 2}, \text{ where } a \text{ is a constant.}$ $\therefore$ The equation of C is $y = x - 1 + \frac{a}{x - 2} = \frac{(x - 1)(x - 2) + a}{x - 2} \Rightarrow y = \frac{x^2 - 3x + 2 + a}{x - 2} \text{ -- (2)}$ Comparing (2) and (1), $p = -3$.
2(ii) [1]	The figure shows a Cartesian coordinate system with a curve labeled 'Curve C'. A vertical dashed line represents the asymptote $x = 2$. A horizontal blue line represents the asymptote $y = \frac{7}{2}$. A dashed line represents the oblique asymptote $y = x - 1$. The curve C has a vertical asymptote at $x = 2$ and an oblique asymptote at $y = x - 1$. The curve intersects the horizontal asymptote $y = \frac{7}{2}$ at two points, with x-coordinates marked as 2.5 and 4. The curve also passes through the point $(0, -\frac{3}{2})$, which is labeled. The origin is marked with 'O'. For $f(x) = \frac{x^2 - 3x + 3}{x - 2} > \frac{7}{2}$, from graph, $2 < x < 2.5$ or $x > 4$.

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Qn	Solution
3(a) [2]	$\int \frac{x}{\sqrt{x^2 - 4}} dx$ $= \frac{1}{2} \int 2x(x^2 - 4)^{-\frac{1}{2}} dx$ $= \frac{1}{2} \left[\frac{(x^2 - 4)^{\frac{1}{2}}}{\frac{1}{2}} \right] + C$ $= \sqrt{x^2 - 4} + C$
3(b) [3]	Let $u = \ln(2x)$ $\frac{dv}{dx} = x^2$ $\frac{du}{dx} = \frac{1}{x} \quad v = \frac{x^3}{3}$ $\int x^2 \ln(2x) dx$ $= \left(\frac{x^3}{3} \right) \ln(2x) - \int \frac{1}{x} \left(\frac{x^3}{3} \right) dx$ $= \frac{1}{3} x^3 \ln(2x) - \frac{1}{3} \int x^2 dx$ $= \frac{1}{3} x^3 \ln(2x) - \frac{1}{9} x^3 + C$
3(c) [2]	$\int \frac{\sin 2x}{1 + \cos^2 x} dx$ $= - \int \frac{-2 \cos x \sin x}{1 + \cos^2 x} dx$ $= - \ln 1 + \cos^2 x + C$ $= - \ln(1 + \cos^2 x) + C$

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Qn	Solution
4(i) [3]	Let x, y and z be the usual selling price of Blend A, B and C coffee beans respectively. $x + y + z = 176 \text{ --- (2)}$ $0.85x + 0.90y + 0.95z = 161.80 \text{ --- (2)}$ $0.93x + 0.95y + 0.98z = 169.56 \text{ --- (3)}$ From GC: Hence $x = \\$32$, $y = \\$44$, $z = \\$100$
4(ii) [3]	Price difference: $169.56 - 161.80 = 7.76$ To make it more attractive: Additional loyalty discount > price difference $\left(\frac{x}{100}\right)(32) + \left(\frac{x}{100}\right)(44) > 7.76$ $x > 10.2$ $x = 11$ Note: $x\% = \frac{x}{100}$

Qn	Solution
5(i) [4]	Let $u = x^2 + 3 \Rightarrow \frac{du}{dx} = 2x$ $\int \frac{x^3}{\sqrt{x^2 + 3}} dx$ $= \int \frac{x^2}{\sqrt{x^2 + 3}} (x dx)$ $= \int \frac{u-3}{\sqrt{u}} \left(\frac{du}{2} \right)$ $= \frac{1}{2} \int u^{\frac{1}{2}} - 3u^{-\frac{1}{2}} du$ $= \frac{1}{2} \left(\frac{u^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} - 3 \frac{u^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} \right) + C$ $= \frac{1}{3} u^{\frac{3}{2}} - 3u^{\frac{1}{2}} + C$ $= \frac{1}{3} \sqrt{(x^2 + 3)^3} - 3\sqrt{x^2 + 3} + C \text{ (shown)}$ Or Let $u = x^2 + 3 \Rightarrow \frac{du}{dx} = 2x$ $\int \frac{x^3}{\sqrt{x^2 + 3}} dx$ $= \int \frac{(u-3)^{\frac{3}{2}}}{\sqrt{u}} \left(\frac{du}{2x} \right)$ Continue to replace x in terms of u if it can not be done easily with 1 step $= \int \frac{(u-3)^{\frac{3}{2}}}{\sqrt{u}} \left(\frac{du}{2(u-3)^{\frac{1}{2}}} \right)$ $= \int \frac{u-3}{\sqrt{u}} \left(\frac{du}{2} \right)$ $= \frac{1}{2} \int u^{\frac{1}{2}} - 3u^{-\frac{1}{2}} du$ $= \frac{1}{2} \left(\frac{u^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} - 3 \frac{u^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} \right) + C$ $= \frac{1}{3} (x^2 + 3)^{\frac{3}{2}} - 3(x^2 + 3)^{\frac{1}{2}} + C \text{ (shown)}$

5(ii)
[3]

$$\int_{-1}^3 \left| \frac{3x^3}{\sqrt{x^2+3}} \right| dx$$

$$= \int_{-1}^0 \left| \frac{3x^3}{\sqrt{x^2+3}} \right| dx + \int_0^3 \left| \frac{3x^3}{\sqrt{x^2+3}} \right| dx$$

$$= \int_{-1}^0 \frac{-3x^3}{\sqrt{x^2+3}} dx + \int_0^3 \frac{3x^3}{\sqrt{x^2+3}} dx$$

[original curve reflected in the x-axis] [Part of original curve]

$$= - \left[\sqrt{(x^2+3)^3} - 9\sqrt{x^2+3} \right]_{-1}^0 + \left[\sqrt{(x^2+3)^3} - 9\sqrt{x^2+3} \right]_0^3$$

$$= - \left[(\sqrt{27} - 9\sqrt{3}) - (\sqrt{64} - 9\sqrt{4}) \right] + \left[(\sqrt{12^3} - 9\sqrt{12}) - (\sqrt{3^3} - 9\sqrt{3}) \right]$$

$$= - \left[(3\sqrt{3} - 9\sqrt{3}) - (8 - 18) \right] + \left[(12\sqrt{12} - 9\sqrt{12}) - (3\sqrt{3} - 9\sqrt{3}) \right]$$

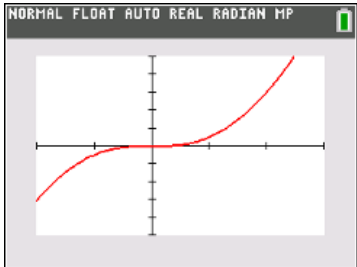
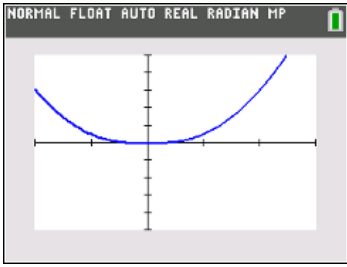
$$= - \left[10 - 6\sqrt{3} \right] + \left[(3\sqrt{12}) + 6\sqrt{3} \right]$$

$$= - \left[10 - 6\sqrt{3} \right] + \left[(3\sqrt{4(3)}) + 6\sqrt{3} \right]$$

$$= - \left[10 - 6\sqrt{3} \right] + \left[6\sqrt{3} + 6\sqrt{3} \right]$$

$$= -10 + 18\sqrt{3}$$

Use GC:

Let $f(x) = \frac{3x^3}{\sqrt{x^2+3}}$	Let $f(x) = \left \frac{3x^3}{\sqrt{x^2+3}} \right $
	
$-1 \leq x < 0$, we have $f(x) < 0$ $0 \leq x \leq 3$, we have $f(x) \geq 0$	$\int_{-1}^0 \frac{-3x^3}{\sqrt{x^2+3}} dx$ [Reflection in x-axis] $+ \int_0^3 \frac{3x^3}{\sqrt{x^2+3}} dx$ [Part of original curve]

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Qn	Solution
6(i) [3]	Area : $S = 2 \times \left[\frac{1}{2} \times \pi \left(\frac{x}{2} \right)^2 \right] + 2xy$ $S = \frac{\pi x^2}{4} + 2xy$ Perimeter: $25 = 2 \times \left[\frac{1}{2} \times 2\pi \left(\frac{x}{2} \right) \right] + 2x + 2y$ $2y = 25 - \pi x - 2x$ $S = \frac{\pi x^2}{4} + (25 - \pi x - 2x)x$ $S = \frac{\pi x^2}{4} + 25x - \pi x^2 - 2x^2$ $S = 25x - 2x^2 - \frac{3}{4}\pi x^2 \quad (\text{shown})$
6(ii) [4]	$S = 25x - 2x^2 - \frac{3}{4}\pi x^2$ $\frac{dS}{dx} = 25 - 4x - \frac{3}{2}\pi x$ At stationary point, $\frac{dS}{dx} = 0$ $25 - 4x - \frac{3}{2}\pi x = 0$ $50 - 8x - 3\pi x = 0$ $(8 + 3\pi)x = 50$ $x = \frac{50}{8 + 3\pi}$ $\frac{dS}{dx} = 25 - 4x - \frac{3}{2}\pi x$ $\frac{d^2S}{dx^2} = -4 - \frac{3}{2}\pi = -8.71 < 0$ S is maximum area

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Qn	Solution																		
7(i) [2]	GP: $a = 500, r = 0.6$ $u_0 = 500$ $u_1 = 0.6 \times 500 = 300$ $u_2 = 0.6 \times (0.6 \times 500) = 500 \times 0.6^2 = 180$ (verified)																		
7 (ii) [2]	<table border="1"><thead><tr><th>Week n</th><th>New case, u_n</th><th>Total Cumulative, S_n</th></tr></thead><tbody><tr><td>0</td><td>500</td><td>500</td></tr><tr><td>1</td><td>500×0.6^1</td><td>$500 + 500 \times 0.6^1$</td></tr><tr><td>2</td><td>500×0.6^2</td><td>$500 + 500 \times 0.6^1 + 500 \times 0.6^2$</td></tr><tr><td>...</td><td>...</td><td>...</td></tr><tr><td>n</td><td>500×0.6^n</td><td>$500 + 500 \times 0.6^1 + 500 \times 0.6^2$ $+ \dots + 500 \times 0.6^n$</td></tr></tbody></table> $S_n = 500 + 500 \times 0.6^1 + 500 \times 0.6^2 + \dots + 500 \times 0.6^n$ $S_n = 500(1 + 0.6 + 0.6^2 + \dots + 0.6^n)$ $S_n = 500 \left[\frac{1(1 - 0.6^{n+1})}{1 - 0.6} \right]$ $S_n = 1250(1 - 0.6^{n+1})$ (shown)	Week n	New case, u_n	Total Cumulative, S_n	0	500	500	1	500×0.6^1	$500 + 500 \times 0.6^1$	2	500×0.6^2	$500 + 500 \times 0.6^1 + 500 \times 0.6^2$	...	...	...	n	500×0.6^n	$500 + 500 \times 0.6^1 + 500 \times 0.6^2$ $+ \dots + 500 \times 0.6^n$
Week n	New case, u_n	Total Cumulative, S_n																	
0	500	500																	
1	500×0.6^1	$500 + 500 \times 0.6^1$																	
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...	...	...																	
n	500×0.6^n	$500 + 500 \times 0.6^1 + 500 \times 0.6^2$ $+ \dots + 500 \times 0.6^n$																	
7 (iii) [1]	Method 1: GC $1250(1 - 0.6^{n+1}) > 1240$ From GC <table border="1"><tbody><tr><td>n</td><td>$1250(1 - 0.6^{n+1})$</td></tr><tr><td>8</td><td>1237.4</td></tr><tr><td>9</td><td>1242.4 > 1240</td></tr></tbody></table> Method 2: Solving Inequality $S_n = 1250(1 - 0.6^{n+1})$ $1250(1 - 0.6^{n+1}) > 1240$ $0.6^{n+1} < 1 - \frac{1240}{1250}$ $n > \frac{\ln\left(\frac{1}{125}\right)}{\ln(0.6)} - 1$ $n > 8.45$ $n_c = 9$ $\ln(0.6) < 0$, handle the inequality sign accordingly	n	$1250(1 - 0.6^{n+1})$	8	1237.4	9	1242.4 > 1240												
n	$1250(1 - 0.6^{n+1})$																		
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7 (iv) [2]	$S_n = 1250(1 - 0.6^{n+1})$ $n \rightarrow \infty, 0.6^{n+1} \rightarrow 0, S_n \rightarrow 1250$ Since the maximum cumulative infection is $1250 < 1280$, the patient capacity will be sufficient.
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Qn	Solution
8 (i) [2]	Method 1 $y = e^{2\sin^{-1}x} \Rightarrow \ln y = 2\sin^{-1}x$ Differentiating both sides wrt x: $\frac{1}{y} \frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$ $\sqrt{1-x^2} \frac{dy}{dx} = 2y \quad (\text{shown})$ Method 2 $y = e^{2\sin^{-1}x}$ $\frac{dy}{dx} = e^{2\sin^{-1}x} \times \frac{2}{\sqrt{1-x^2}}$ $\sqrt{1-x^2} \frac{dy}{dx} = 2e^{2\sin^{-1}x}$ $\sqrt{1-x^2} \frac{dy}{dx} = 2y \quad (\text{shown})$
8 (ii) [4]	$(1-x^2) \left(\frac{dy}{dx} \right)^2 = 4y^2$ Differentiating both sides wrt x: $(1-x^2) \times 2 \left(\frac{dy}{dx} \right) \left(\frac{d^2y}{dx^2} \right) + (-2x) \left(\frac{dy}{dx} \right)^2 = 8y \frac{dy}{dx}$ ----- <i>Alternatively,</i> $(1-x^2)^{\frac{1}{2}} \frac{dy}{dx} = 2y$ Differentiating both sides wrt x: $(1-x^2)^{\frac{1}{2}} \frac{d^2y}{dx^2} + \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x) \frac{dy}{dx} = 2 \frac{dy}{dx}$ -----

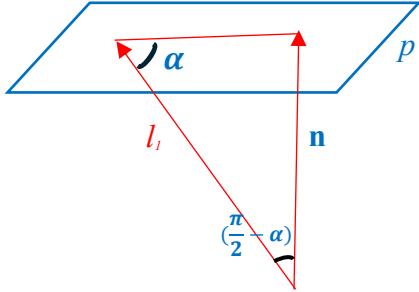
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	When $x = 0$, $y = e^{2\sin^{-1}0} \Rightarrow y = 1$ $\sqrt{1-0^2} \frac{dy}{dx} = 2(1) \Rightarrow \frac{dy}{dx} = 2$ $(1-0^2) \times 2(2) \left(\frac{d^2y}{dx^2} \right) + (0)(2)^2 = 8(1)(2) \Rightarrow \frac{d^2y}{dx^2} = 4$ <i>Alternatively,</i> $(1-0^2)^{\frac{1}{2}} \frac{d^2y}{dx^2} + \frac{1}{2}(1-0^2)^{-\frac{1}{2}}(0)(2) = 2(2) \Rightarrow \frac{d^2y}{dx^2} = 4$ $\therefore$ The Maclaurin series for y: $y = 1 + 2x + 4\left(\frac{x^2}{2!}\right) + \dots$ $\Rightarrow y = 1 + 2x + 2x^2 + \dots$
8 (iii) [2]	$\int_0^{0.1} x e^{2\sin^{-1}x} dx \approx \int_0^{0.1} x(1 + 2x + 2x^2) dx = 0.00571667$ $= 0.005717 \text{ (4 s.f.)}$

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Qn	Solution
9 (i) [2]	$\overrightarrow{AB}$ is parallel to $\begin{pmatrix} k \\ 1 \\ 2 \end{pmatrix}$ $\overrightarrow{OB} - \overrightarrow{OA} = \lambda \begin{pmatrix} k \\ 1 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 17 \\ 10 \\ 19 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \lambda \begin{pmatrix} k \\ 1 \\ 2 \end{pmatrix}$ $\lambda \begin{pmatrix} k \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 16 \end{pmatrix} = 8 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ By comparison: when $\lambda = 8$, $k = 2$ (shown)
9 (ii) [4]	Observe the l_1 and l_2 are not parallel. l_1 and l_2 either are skewed lines or intersect at a point. We need to further check if the lines intersect. $l_1: \vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}$ $l_2: \vec{r} = \begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \mu \in \mathbb{R}$ Method 1: Use GC $\begin{pmatrix} 1+2\lambda \\ 2+\lambda \\ 3+2\lambda \end{pmatrix} = \begin{pmatrix} 3+2\mu \\ 4 \\ 6+\mu \end{pmatrix}$ $2\lambda - 2\mu = 2$ $\lambda = 2$ $2\lambda - \mu = 3$ From GC, $\lambda = 2$ and $\mu = 1$. The system of eqns are consistent and solution is unique.

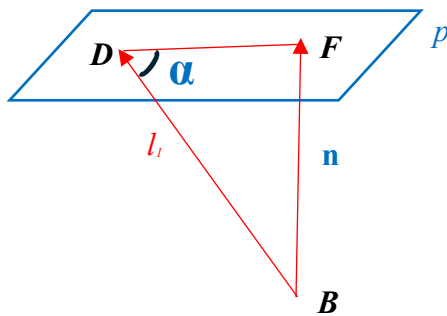
	Both lines intersect at position vector, The lines intersect. Sub $\mu = 1$ into l_2. $\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} + (1) \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix}$ Method 2: Solve manually If both lines intersect at point, there is an unique solution $\begin{pmatrix} 1+2\lambda \\ 2+\lambda \\ 3+2\lambda \end{pmatrix} = \begin{pmatrix} 3+2\mu \\ 4 \\ 6+\mu \end{pmatrix}$ Only 1 unknown to start with! Compare j component $2 + \lambda = 4$ $\lambda = 2$ Compare i component $1 + 2(2) = 3 + 2\mu$ $\mu = 1$ Note: To check if the system is consistent When $\lambda = 2$ and $\mu = 1$, $\text{LHS} = \begin{pmatrix} 1+2[2] \\ 2+[2] \\ 3+2[2] \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 3+2[1] \\ 4 \\ 6+[1] \end{pmatrix} = \text{RHS}$ Both lines intersect at position vector, $\begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix}$
9(iii) [We 3]	Given α the acute angle between l_1 and p (refer to diagram below) $\cos\left(\frac{\pi}{2} - \alpha\right) = \frac{\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}{\left(\sqrt{2^2 + 1^2 + 2^2}\right)\left(\sqrt{1^2 + 0^2 + 1^2}\right)}$ $\sin \alpha = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3}$

	$\sin^2 \alpha + \cos^2 \alpha = 1$ $\cos \alpha = \sqrt{1 - \sin^2 \alpha} = \sqrt{1 - \left(\frac{2\sqrt{2}}{3}\right)^2} = \frac{1}{3}$ 
9 (iv) [3]	<u>Method 1: Intersection of a Line and a Plane</u> Point F, $\overrightarrow{OF}$ lies on ℓ_{BF} : $\vec{r} = \begin{pmatrix} 17 \\ 10 \\ 19 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \text{ for some } \alpha \in \mathbb{R} \text{ --- (1)}$ Sub (1) into the eqn of plane p as Point F is also on plane p : $\begin{pmatrix} 17 + \alpha \\ 10 \\ 19 + \alpha \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 16$ $17 + \alpha + 19 + \alpha = 16$ $\alpha = -\frac{20}{2} = -10$ $\overrightarrow{OF} = \overrightarrow{OB} + \overrightarrow{BF}$ $\overrightarrow{OF} = \begin{pmatrix} 17 \\ 10 \\ 19 \end{pmatrix} + (-10) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \\ 9 \end{pmatrix}$ Coordinates of point F is $(7, 10, 9)$. <u>Method 2: Find $\overrightarrow{BF}$ by projecting $\overrightarrow{BD}$ onto $\mathbf{n}$, where D is a point on plane p.</u> Obtain a specific point, D from plane p, $\overrightarrow{OD} = \begin{pmatrix} 16 \\ 0 \\ 0 \end{pmatrix}$.

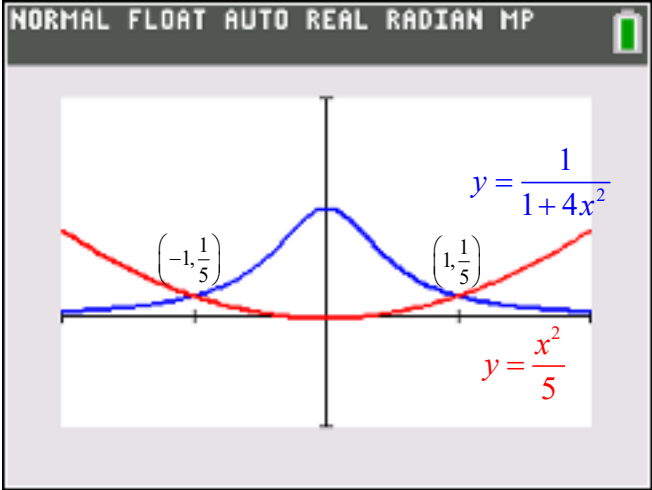
$$\begin{aligned}\overrightarrow{BF} &= (\overrightarrow{BD} \cdot \underline{n}) \underline{n} \\ &= \frac{\left\{ \begin{pmatrix} 16 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 17 \\ 10 \\ 19 \end{pmatrix} \right\} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}{|\sqrt{2}|} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \\ &= \frac{-20}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = -10 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\end{aligned}$$

$$\overrightarrow{OF} = \overrightarrow{OB} + \overrightarrow{BF}$$

$$\overrightarrow{OF} = \begin{pmatrix} 17 \\ 10 \\ 19 \end{pmatrix} + (-10) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \\ 9 \end{pmatrix}$$



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Qn	Solution
10(i) [2]	$y = \frac{1}{1+4x^2} \text{----- (1)}$ $y = \frac{x^2}{5} \text{----- (2)}$ Using GC: The intersection points are $\left(-1, \frac{1}{5}\right), \left(1, \frac{1}{5}\right)$. 
10(ii) [5]	Exact area of region R $= \int_{-1}^1 \left(\frac{1}{1+4x^2} - \frac{x^2}{5} \right) dx$ $= \int_{-1}^1 \frac{1}{4} \left(\frac{1}{\left(\frac{1}{2}\right)^2 + x^2} \right) - \left(\frac{x^2}{5} \right) dx$ $= \left[\frac{1}{4} \left[\frac{1}{\left(\frac{1}{2}\right)} \tan^{-1} \left(\frac{x}{\frac{1}{2}} \right) \right] - \frac{1}{15} x^3 \right]_{-1}^1$ $= \left[\left[\frac{1}{2} \tan^{-1}(2x) \right] - \frac{1}{15} x^3 \right]_{-1}^1$ $= \left[\left(\frac{1}{2} \tan^{-1}(2) - \frac{1}{15} (1)^3 \right) - \left(\frac{1}{2} \tan^{-1}(-2) - \frac{1}{15} (-1)^3 \right) \right]$ $= \frac{1}{2} \tan^{-1}(2) - \frac{1}{15} + \frac{1}{2} \tan^{-1}(2) - \frac{1}{15}$ $= \tan^{-1}(2) - \frac{2}{15} \text{ units}^2$

10(iii)
[5]

$$y = \frac{x^2}{5} \Rightarrow x_1^2 = 5y$$

$$y = \frac{1}{1+4x^2} \Rightarrow x_2^2 = \frac{1}{4} \left(\frac{1}{y} - 1 \right)$$

Exact volume generated

$$= \pi \int_0^{\frac{1}{5}} x_1^2 \, dy + \pi \int_{\frac{1}{5}}^1 x_2^2 \, dy$$

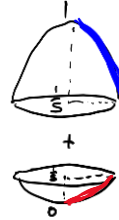
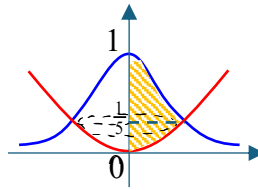
$$= \pi \int_0^{\frac{1}{5}} 5y \, dy + \pi \int_{\frac{1}{5}}^1 \frac{1}{4} \left(\frac{1}{y} - 1 \right) \, dy$$

$$= \pi \left[\frac{5}{2} y^2 \right]_0^{\frac{1}{5}} + \frac{\pi}{4} [\ln y - y]_{\frac{1}{5}}^1$$

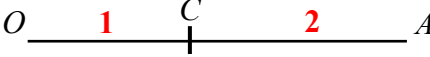
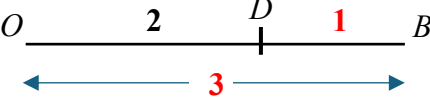
$$= \pi \left(\frac{1}{10} - 0 \right) + \frac{\pi}{4} \left[(\ln(1) - 1) - \left(\ln\left(\frac{1}{5}\right) - \frac{1}{5} \right) \right]$$

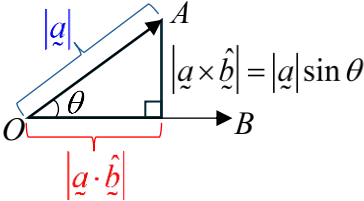
$$= \frac{\pi}{10} + \frac{\pi}{4} \left[-\frac{4}{5} - \ln\left(\frac{1}{5}\right) \right]$$

$$= \frac{\pi}{4} \ln 5 - \frac{\pi}{10} \text{ unit}^3$$



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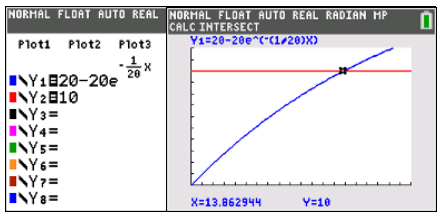
Qn	Solution
11(i) [2]	$\overrightarrow{OC} = \frac{1}{3} \overrightarrow{OA}$  $\overrightarrow{OC} = \frac{1}{3} \underline{a}$ $\overrightarrow{OD} = \frac{2}{3} \overrightarrow{OB}$  $\overrightarrow{OD} = \frac{2}{3} \underline{b}$
11(ii) [5]	$\overrightarrow{OE} = \underline{r} = \overrightarrow{OA} + \lambda \overrightarrow{AD} \text{ --- (1)}$ $\overrightarrow{OE} = \underline{r} = \overrightarrow{OB} + \mu \overrightarrow{BC} \text{ --- (2)}$ Solve (1) = (2): $\Rightarrow \overrightarrow{OA} + \lambda \overrightarrow{AD} = \overrightarrow{OB} + \mu \overrightarrow{BC}$ $\overrightarrow{OA} + \lambda (\overrightarrow{OD} - \overrightarrow{OA}) = \overrightarrow{OB} + \mu (\overrightarrow{OC} - \overrightarrow{OB})$ $\underline{a} + \lambda \left(\frac{2}{3} \underline{b} - \underline{a} \right) = \underline{b} + \mu \left(\frac{1}{3} \underline{a} - \underline{b} \right)$ $(1 - \lambda) \underline{a} + \frac{2}{3} \lambda \underline{b} = \frac{1}{3} \mu \underline{a} + (1 - \mu) \underline{b}$ Comparing coefficient of $\underline{a}$: $1 - \lambda = \frac{1}{3} \mu$ $\Rightarrow \lambda + \frac{1}{3} \mu = 1 \text{ --- (3)}$ Comparing coefficient of $\underline{b}$: $\frac{2}{3} \lambda = 1 - \mu$ $\Rightarrow \frac{2}{3} \lambda + \mu = 1 \text{ --- (4)}$ Solving (3) & (4), by GC: $\lambda = \frac{6}{7}, \mu = \frac{3}{7}$ Sub $\lambda = \frac{6}{7}$ into (1): $\begin{aligned} \overrightarrow{OE} &= \underline{a} + \frac{6}{7} \left(\frac{2}{3} \underline{b} - \underline{a} \right) \\ &= \underline{a} + \frac{4}{7} \underline{b} - \frac{6}{7} \underline{a} \\ &= \frac{1}{7} \underline{a} + \frac{4}{7} \underline{b} \text{ (shown)} \end{aligned}$

11 (iii) [3]	Area of $\triangle OAE = \frac{1}{2} \vec{OA} \times \vec{OE}$ $= \frac{1}{2} \left \vec{a} \times \left(\frac{1}{7} \vec{a} + \frac{4}{7} \vec{b} \right) \right $ $= \frac{1}{2} \left \left(\vec{a} \times \frac{1}{7} \vec{a} \right) + \left(\vec{a} \times \frac{4}{7} \vec{b} \right) \right $ $= \frac{1}{2} \left \frac{1}{7} (\vec{a} \times \vec{a}) + \frac{4}{7} (\vec{a} \times \vec{b}) \right $ $= \frac{1}{2} \left 0 + \frac{4}{7} (\vec{a} \times \vec{b}) \right \quad (\because \vec{a} \times \vec{a} = 0)$ $= \frac{2}{7} \vec{a} \times \vec{b} , \text{ where } k = \frac{2}{7} \text{ (shown).}$
11 (iv) [1]	The length of projection of OA onto the line OB.
11(v) [3]	Method 1 Perpendicular distance from A to OB $= \sqrt{(\vec{a})^2 - (\vec{a} \cdot \hat{b})^2}$ $= \sqrt{(\vec{a})^2 - \left(\frac{1}{2} \vec{a} \right)^2} \quad (\vec{a} \cdot \hat{b} = \vec{a} \cdot \vec{b} = \frac{1}{2} \vec{a} \text{ (given)})$ $= \sqrt{ \vec{a} ^2 - \frac{1}{4} \vec{a} ^2}$ $= \sqrt{\frac{3}{4} \vec{a} ^2}$ $= \frac{\sqrt{3}}{2} \vec{a} $  Method 2 Perpendicular distance = $\vec{a} \sin \theta$ (need to find θ) From $\vec{a} \cdot \vec{b} = \frac{1}{2} \vec{a}$, we have $\vec{a} \vec{b} \cos \theta = \frac{1}{2} \vec{a}$ $\Rightarrow \cos \theta = \frac{1}{2} \text{ (since } \vec{b} = 1)$ $\Rightarrow \theta = 60^\circ$ Perpendicular distance = $\vec{a} \sin 60^\circ = \frac{\sqrt{3}}{2} \vec{a}$

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Qn	Solution
12 (i) [1]	At any point in time, the inflow rate is equal to the outflow rate (both 5 litres per minute). Therefore, the volume of the mixture in the tank remains constant.
12 (ii) [1]	Salt inflow rate = Salt concentration of brine $\times$ Brine inflow rate $\frac{dQ_{\text{in}}}{dt} = \frac{dQ_{\text{in}}}{dV} \times \frac{dV}{dt}$ $= 0.2 \times 5$ $= 1 \text{ kg / min (shown)}$
12 (iii) [2]	Mass of salt in tank at time t is Q kg, Vol of mixture = 100 litres Salt concentration in tank at time $t = \frac{Q}{100}$ kg/litre Salt outflow rate = Salt Concentration in tank $\times$ Outflow rate of mixture $\frac{dQ_{\text{out}}}{dt} = \frac{dQ_{\text{out}}}{dV} \times \frac{dV}{dt}$ $= \frac{Q}{100} \times 5$ $= \frac{Q}{20}$ $\frac{dQ}{dt} = \frac{dQ_{\text{in}}}{dt} - \frac{dQ_{\text{out}}}{dt} = 1 - \frac{Q}{20} \text{ (shown)}$
12 (iv) [5]	$\frac{dQ}{dt} = 1 - \frac{Q}{20} = \frac{20 - Q}{20}$ $\int \frac{1}{20 - Q} dQ = \int \frac{1}{20} dt$ $-\ln 20 - Q = \frac{1}{20}t + c$ $\ln 20 - Q = -\frac{t}{20} - c$ $ 20 - Q = e^{-\frac{t}{20} - c}$ $20 - Q = Ae^{-\frac{t}{20}}$ When $t = 0$, $Q = 0$ (initially pure water). $20 - 0 = Ae^0 \Rightarrow A = 20$ $\therefore Q = 20 - 20e^{-\frac{t}{20}}$

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12 (v) [2]	Salt concentration in tank = 0.1 kg/litre Mass of salt = $0.1 \times 100 = 10$ kg Sub $Q = 10$: $10 = 20 - 20e^{-\frac{t}{20}}$ Method 1 (GC Graph) From GC, $t = 13.863$ $t = 13.9$ (3 s.f.)  Method 2 (Algebraic) $10 = 20 - 20e^{-\frac{t}{20}}$ $e^{-\frac{t}{20}} = \frac{1}{2}$ $-\frac{t}{20} = \ln \frac{1}{2}$ $t = -20 \ln \frac{1}{2} = 13.9$ (3 s.f.)
12 (vi) [1]	$Q = 20 - 20e^{-\frac{t}{20}}$ When $t \rightarrow \infty$, $e^{-\frac{t}{20}} \rightarrow 0$, $Q \rightarrow 20$ Q increases and approaches 20. 